A tour of paged media

A short catalogue of the things HTML-to-PDF tools put into PDF files, made to give a PDF reader plenty to do: vector graphics, rounded and patterned borders, counters, footnotes, bookmarks, internal and external links, and, added afterwards by Broadside, optional content layers you can switch on and off.

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PROOF · cover · full bleed

I. Vector graphics

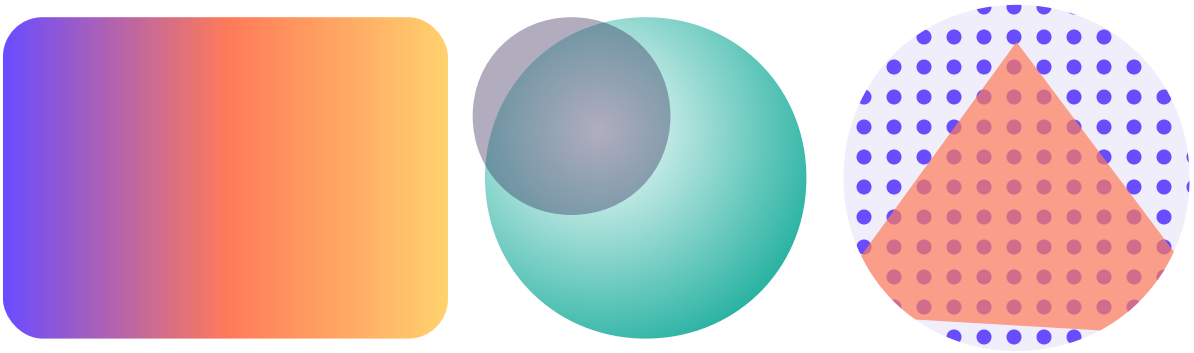
Every illustration in this section is inline SVG, which the producer converts to PDF paths, gradients and clipping rather than to pictures. A good PDF reader draws them sharp at any zoom.

I.1 Paths, strokes and joins



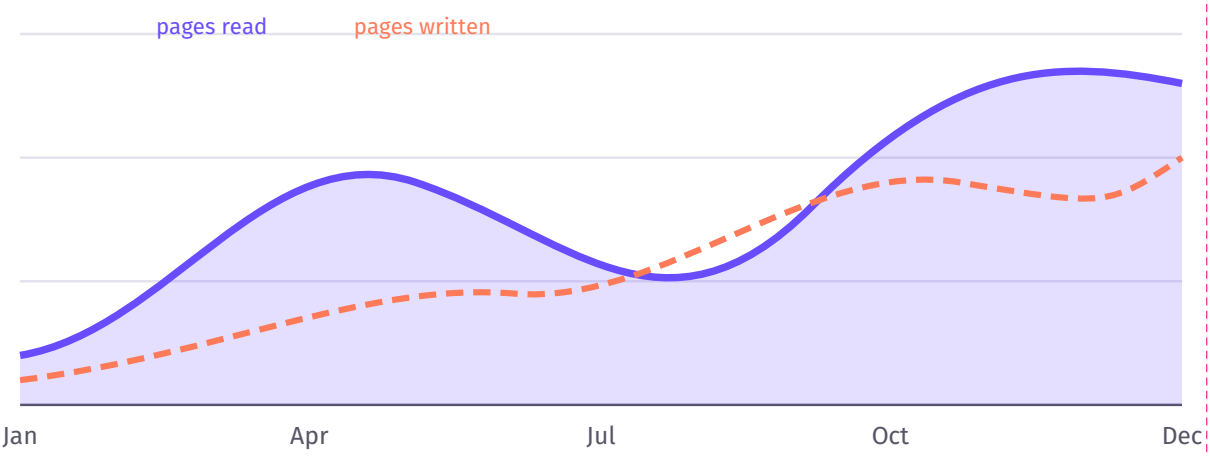
Line joins and caps, each drawn as a single stroked path.

I.2 Gradients, opacity and clipping



A linear gradient, a radial gradient under a translucent circle, and a tiling pattern clipped to a circle.

I.3 A chart

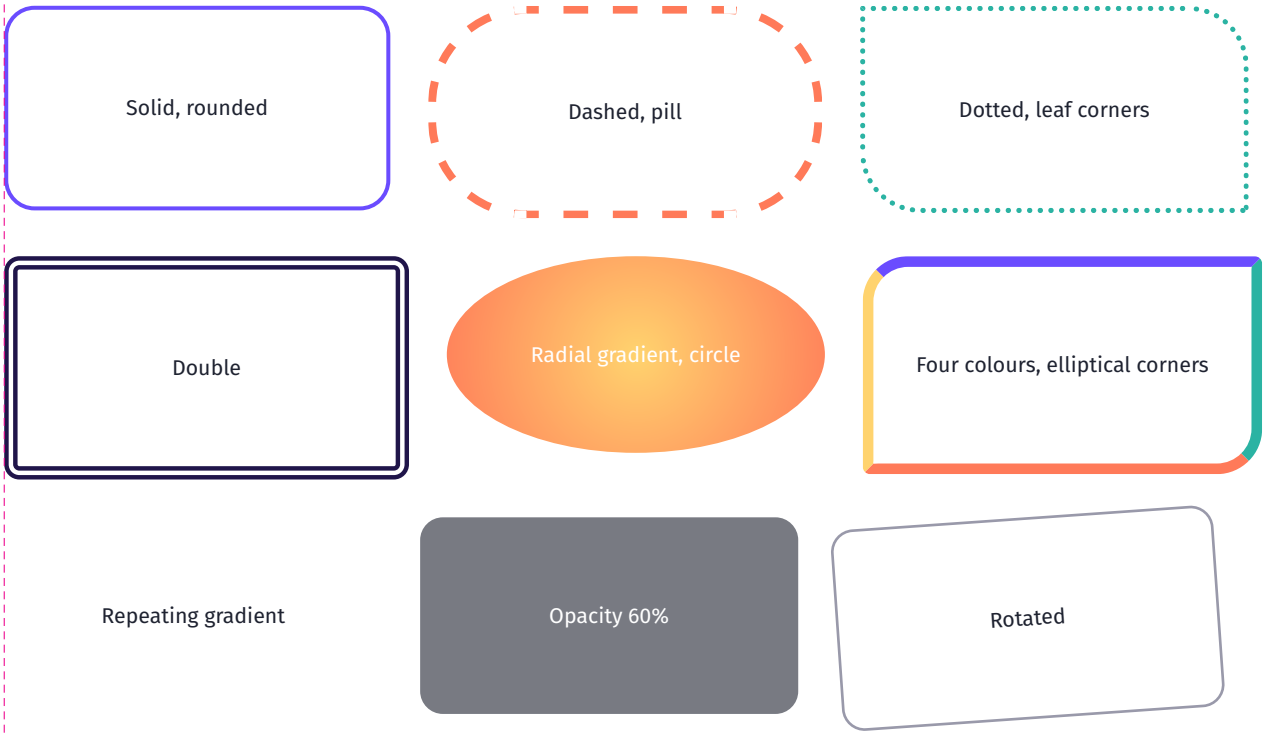


Smooth curves with a filled area beneath and a dashed comparison line (invented data).

PROOF · page 1 · margins 22 / 20 / 24 / 20 mm

II. Borders and backgrounds

CSS borders become stroked or filled paths in PDF. Rounded corners need Bézier curves, dashed and dotted borders need dash arrays that fit the corners, and double borders are two paths. Each box below shows one combination.



> Backgrounds can be drawn inside or under the border; this producer paints them to the outer border edge, so translucent borders show the background through them.[a]

a. Look closely at the dashed pill: its background is white, so the effect is invisible there, but the double border shows a thin line of background between its two strokes.

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III. Counters and lists

Counters number headings, list items, figures and footnotes. This page uses five counter styles; the section numbers above come from a counter too.

III.1 Numbered steps

1. Write the document in HTML and CSS.
2. Let the producer break it into pages.
3. Check the result in more than one reader.
4. Fix the CSS, not the PDF.

III.2 Roman and Greek

I. Preface
II. Introduction
III. Method
IV. Results

α. alpha release
β. beta release
γ. gamma ray

III.3 Checklist

- ✓ Fonts embedded
- ✓ Images at print resolution
- ✓ Links tested
- ✓ Bookmarks present

III.4 Feature table

Feature	In this file	How it is drawn
Inline SVG	Yes	Paths, shadings, patterns, clipping
Rounded borders	Yes	Bézier curves, dash arrays
Counters	Yes	Text generated at layout time
Footnotes	Yes	Moved to the foot of the page
Bookmarks	Yes	The document outline, from headings
Links	Yes	Link annotations, internal and external
Optional content	Added later	Two layers, written by Broadside in an incremental update

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IV. Notes, links and layers

Footnotes are numbered by a counter of their own[b] and placed at the foot of the page by the producer. Internal links jump within the file: back to the vector graphics or the counters. External links leave it, for example to the CSS Paged Media specification ↗ or the Broadside repository ↗.

The bookmarks panel of your reader shows this document's outline: four parts and their sections, built from the headings.

IV.1 Layers

The producer of this page knows nothing about optional content, so Broadside adds it afterwards, in an incremental update that leaves the original file intact. It writes two layers: **Proof marks**, shown by default, which outlines the page margins on every page, and **Draft stamp**, hidden by default, which prints a diagonal “Draft” across each page. Open the layers panel in a reader that supports them and switch them on and off.

> If your reader has no layers panel, the proof marks are simply part of the page, and the draft stamp is never shown. Both behaviours are correct.

b. This is footnote b. The counter restarts on no page; it runs through the whole document.

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